

INDUSTRIAL ACOUSTIC ANOMALY DETECTION: A COMPARATIVE ANALYSIS OF SUPERVISED AND UNSUPERVISED LEARNING PARADIGMS

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ABSTRACT

Acoustic anomaly detection has emerged as a pivotal technology for non-invasive predictive maintenance in Industry 4.0. However, the scarcity of real-world anomalous recordings poses a major challenge for traditional classification. This paper presents a rigorous comparative analysis between an Unsupervised Convolutional Autoencoder (CAE) and a Supervised CNN Classifier using the MIMII industrial fan dataset. Both paradigms are evaluated under baseline conditions (6dB SNR), noise-hostile environments (0dB SNR), and under a zero-shot generalization test on an unseen machine physical unit (ID 06). Experimental results demonstrate that while the supervised CNN achieves a perfect baseline (1.00 AUC) and exhibits high noise resilience, it suffers from catastrophic class inversion (0.21 AUC) on the unseen machine due to shortcut learning. Conversely, the unsupervised CAE, despite experiencing a performance drop under heavy noise (0.71 AUC), maintains logical stability (0.48 AUC) on the unseen domain, highlighting the critical engineering trade-off between local precision and global generalization.

Index Terms— Anomaly Detection, Convolutional Autoencoder, Domain Shift, Shortcut Learning, Predictive Maintenance

INTRODUCTION

In modern manufacturing, non-invasive predictive maintenance is key to reducing operational downtime and preventing catastrophic failures. Traditional vibration-based analysis, while accurate, often requires expensive sensor installations. Acoustic monitoring via microphones offers a highly flexible, cost-effective alternative.

The fundamental scientific challenge in acoustic anomaly detection is the extreme scarcity of failure data [1]. While healthy operating sounds are abundant, machines rarely fail, and when they do, the acoustic signature of the fault is highly variable. Consequently, the unsupervised paradigm, particularly reconstruction-based Autoencoders, is the standard industrial choice, as they learn exclusively from “normal” operating data [1].

However, a recurring question in system design is how these models perform under domain-shifted conditions [2]—when the acoustic characteristics differ between training (source domain) and monitoring (target domain). This study conducts a systematic,

comparative evaluation of unsupervised and supervised paradigms, validating their performance under different noise levels (6dB and 0dB SNR) and testing their generalization limits on an entirely unseen machine domain, exposing the vulnerabilities of traditional local supervision.

DATA ENGINEERING & SIGNAL CONDITIONING

Robust feature extraction is the cornerstone of successful acoustic analysis. We utilize the MIMII dataset (industrial fan category) [1], which captures actual machine operation blended with real factory background noise.

Audio Specifications & Segmentation

The raw database contains 1-minute single-channel PCM .wav recordings sampled at 16,000 Hz. To optimize training efficiency and model responsiveness, each 60-second audio is segmented into non-overlapping windows of **10 seconds**. This data engineering step multiplies our sample size by six while focusing the network on stable, short-term spectral characteristics.

Dataset Distribution & Metadata

Our study focuses specifically on the **industrial fan** category of the MIMII dataset [1]. To ensure absolute transparency and reproducibility, Table 1 details the exact number of 10-second segments available for each machine physical unit used in our training (IDs 00, 02, 04) and generalization test (ID 06) phases. Note that for each machine unit, these segments are mixed with factory background noise to generate separate, parallel datasets at 6dB and 0dB SNR, respectively.

Machine ID	Normal Segments	Anomalous Segments	Total Segments
ID 00	1,011	407	1,418
ID 02	1,016	359	1,375
ID 04	1,033	348	1,381
ID 06 (Unseen Domain)	1,015	361	1,376
Total fleet	4,075	1,475	5,550

Table 1: Fleet distribution of 10-second acoustic segments for industrial fans in our experimental setup [1].

Log-Mel Spectrogram Pipeline

Each 10-second segment is transformed into a 2D time-frequency representation through the following Digital Signal Processing (DSP) pipeline:

1. **Short-Time Fourier Transform (STFT):** We apply a Hamming window of 1024 samples with a hop length of 512 samples (50% overlap), yielding a temporal resolution of 313 frames.
2. **Mel-Filterbank Mapping:** The linear frequency binning is compressed into 128 Mel bands, centering model attention on the lower and mid-frequency regions where fan mechanical signatures (blade pass frequency, bearing friction) reside.
3. **Log-Amplitude Scaling:** The power spectrogram is converted to decibels (dB) using a log-scale to compress the audio's massive range of energy:

$$S_{\log} = 10 \cdot \log_{10}(S_{\text{mel}})$$

Frequency-wise Z-Score Normalization

To isolate the stable machine harmonics from stationary factory background noise, we apply a frequency-wise Z-score normalization. Each of the 128 Mel bands is normalized independently across the temporal dimension:

$$z_{f,t} = \frac{x_{f,t} - \mu_f}{\sigma_f + \varepsilon}$$

Where μ_f and σ_f are the mean and standard deviation of band f calculated globally over the normal training set. This ensures that the network is insensitive to absolute energy biases, focusing instead on structural spectral variations. The final input to the models is a tensor of shape (1, 128, 313).

PROPOSED METHODOLOGIES

We design two distinct architectures to compare unsupervised and supervised paradigms under a controlled, fair experimental setup.

Unsupervised Convolutional Autoencoder (CAE)

The CAE acts as a “familiarity filter.” It is trained exclusively on normal data to reconstruct the input. The network comprises:

- **Encoder:** 4 Conv2D layers with a 3x3 kernel and a stride of 2. Channels increase from 1 to 128, downsampling the input from (1, 128, 313) to a bottleneck of (128, 8, 20). This represents a $16 \times$ compression in frequency, a $15.65 \times$ compression in the temporal dimension, and an overall data parameter compression of $1.95 \times$ due to the channel expansion.
- **Bottleneck:** A Dropout layer (0.2) is applied to prevent the network from memorizing noise.
- **Decoder:** 4 ConvTranspose2D layers that mirror the encoder.
- **Cropping:** A deterministic crop trims the reconstructed tensor back to (128, 313).

During inference, the reconstruction error (MSE) is calculated per temporal frame. The decision metric, or Anomaly Score, is defined as the **90th percentile** of these frame-wise errors:

$$\text{Score} = P_{90}(\{\text{MSE}_t\}_{t=1}^{313})$$

This formulation makes the detector robust against short, spurious factory noises.

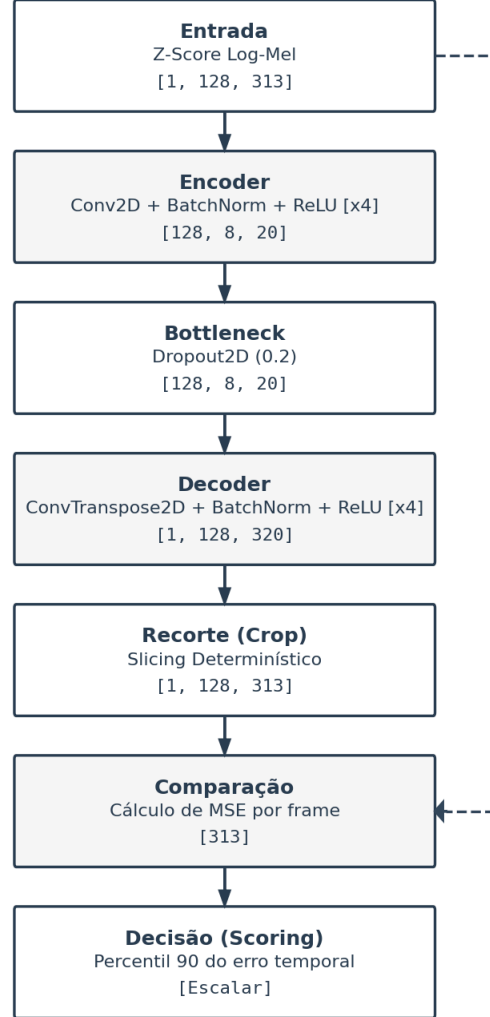


Figure 1: Operational flowchart of the proposed Unsupervised Convolutional Autoencoder (CAE) architecture.

Supervised CNN Classifier

To evaluate the absolute representation capability of the feature space, we implement a supervised CNN classifier. To ensure a strictly fair comparison, the CNN uses the **exact same Encoder architecture** as the CAE (but trained entirely from scratch using labels), followed by:

- A **Flatten** layer.
- A fully-connected neural network (**FC Head**) reducing 20,480 features to 512, followed by a classification neuron.
- A **Sigmoid** activation function outputting a failure probability (P_{anormal}) between 0.0 and 1.0.

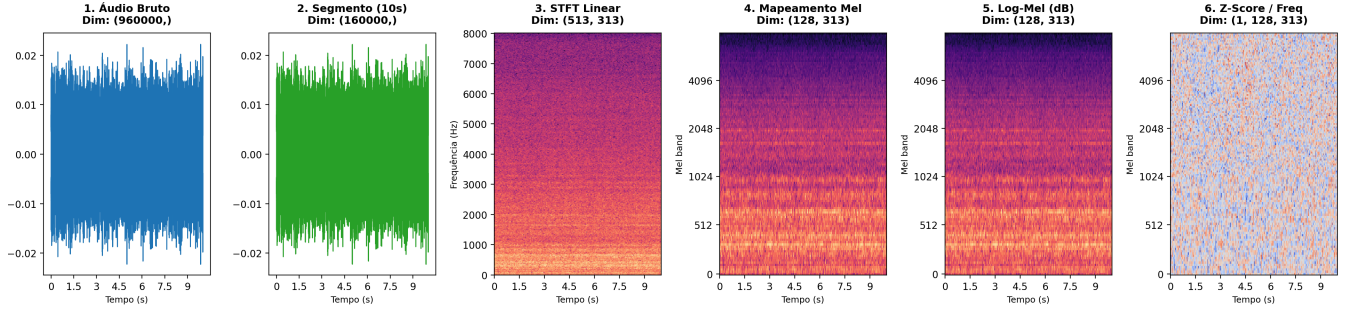


Figure 3: Step-by-step 1D/2D digital signal conditioning and normalization sequence of a 10-second industrial fan acoustic recording.

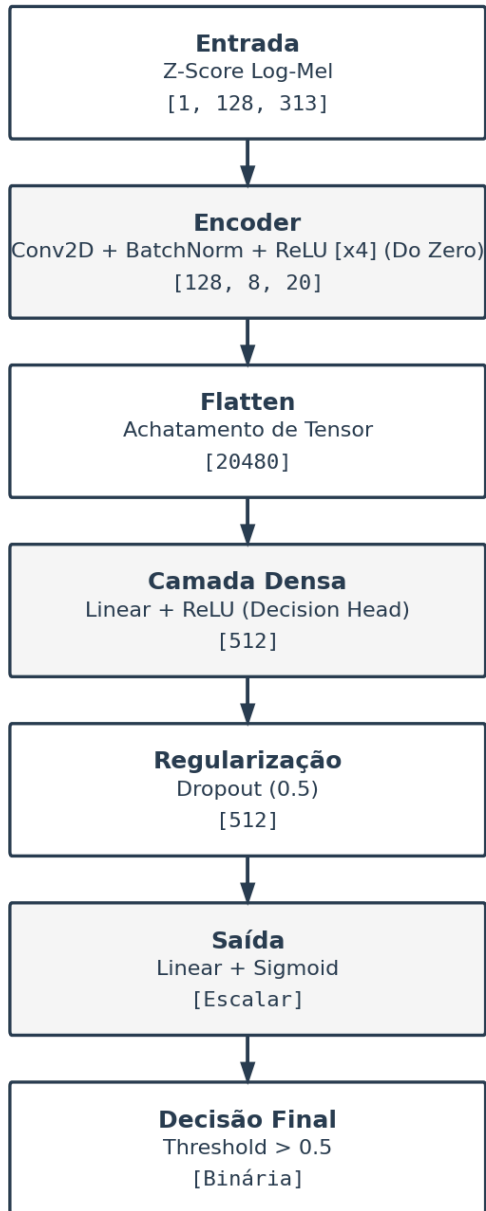


Figure 2: Operational flowchart of the comparative Supervised CNN Classifier architecture.

Model Training & Optimization

To guarantee a reproducible and scientifically rigorous comparative framework, both networks are optimized under a standardized training regimen:

- **Unsupervised CAE:** The network is trained for 50 epochs with a batch size of 32. We employ the **Adam** optimizer with a learning rate of $\eta = 10^{-3}$. The optimization objective is strictly to minimize the reconstruction MSE loss over the “normal” operational subset of IDs 00, 02, and 04.
- **Supervised CNN:** The classifier is trained for 30 epochs with a batch size of 32 (early convergence observed around epoch 14). We use the **Adam** optimizer with a conservative learning rate of $\eta = 10^{-4}$ to prevent premature probability saturation of the Sigmoid activation and ensure smooth convergence of the BCE loss.
- **Framework & Initialization:** Models are implemented in PyTorch, initialized with random weights, and evaluated using an 80/20 train-validation split stratified by class labels.

EXPERIMENTAL RESULTS

Both models were trained on data from Machine IDs 00, 02, and 04 under a 6dB SNR. Testing was conducted on independent baseline samples, 0dB noise stress test samples, and an unseen machine unit (ID 06).

Baseline Performance (6dB SNR)

Under controlled conditions, the specialized “Mono-ID” CAE (trained and tested only on ID 02) is implemented to replicate the benchmark conditions of the original MIMII dataset paper [1]. It achieves an outstanding **0.99 ROC-AUC (0.92 p-AUC)**, successfully reproducing the exact **0.99 AUC** reported by Purohit et al. [1] (Table 3, Fan ID 02, 6dB SNR) and validating the technical integrity of our signal processing pipeline.

However, we introduce a crucial evaluation step: we scale the training to a unified, collective “Multi-ID” CAE trained jointly on IDs 00, 02, and 04. Under these conditions, the CAE’s performance drops to **0.82 ROC-AUC (0.71 p-AUC)**. This gap represents the “Cost of Scalability.” Under identical Multi-ID conditions, the supervised CNN achieves a perfect **1.00 ROC-AUC**.

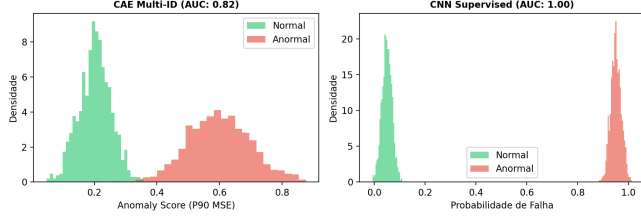


Figure 4: Baseline performance distributions at 6dB SNR, showcasing the CNN’s perfect separation versus the CAE’s cost of scalability.

Noise Resilience Stress Test (0dB SNR)

When background noise is elevated to 0dB (noise as loud as the machine), performance degrades:

- The supervised CNN shows extreme resilience, dropping only slightly to **0.98 ROC-AUC**.
- The unsupervised CAE drops significantly to **0.71 ROC-AUC**.

A histogram analysis reveals a **noise masking phenomenon**: the high energy of the factory noise increases the reconstruction error of normal samples, shifting their distribution to the right and causing severe overlap with the anomalous distribution.

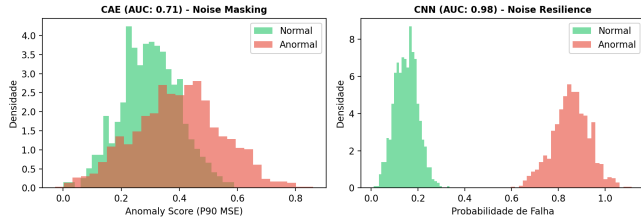


Figure 5: Stress test distributions at 0dB SNR. The CAE suffers from noise masking, while the CNN retains high resilience.

Generalization on Unseen Domain (Machine ID 06)

This blind test reveals the fundamental flaw of local supervision under domain shift conditions. While our study strictly utilizes the original MIMII dataset [1] for all experiments, we frame this particular cross-machine evaluation using the conceptual observations introduced in the MIMII DUE paper [2]. Tanabe et al. [2] demonstrate that different physical units of the exact same machine type (even under normal operation) possess distinct acoustic signatures due to room acoustics, minor manufacturing variances, or installation biases—representing a formal physical domain shift.

In this context, we define our training machines (IDs 00, 02, 04) as the **Source Domain**, and the unseen physical unit (ID 06) as our **Target Domain**. The evaluation on this Target Domain reveals:

- The unsupervised CAE exhibits a stable, though degraded, performance of **0.48 ROC-AUC**, maintaining its underlying logical structure.
- The supervised CNN suffers a **catastrophic collapse to 0.21 ROC-AUC**.

An audit of the CNN’s score distribution reveals a complete **class inversion**: the normal sounds of ID 06 are classified as anomalies with high probability, while actual faults are classified as normal.

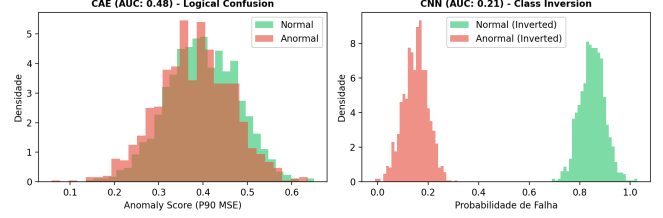


Figure 6: Generalization test on unseen ID 06. The CNN exhibits catastrophic class inversion (Shortcut Learning), whereas the CAE remains logically consistent despite performance degradation.

Evaluation Scenario	CAE (Mono-ID)	CAE (Multi-ID)	CNN (Supervised)
Baseline (6dB SNR)	0.99	0.82	1.00
Stress Test (0dB SNR)	-	0.71	0.98
Unseen Domain (ID 06)	-	0.48	0.21

Table 2: Comparative summary of ROC-AUC performance across noise and domain shifts.

DISCUSSION & TRADE-OFFS

The experimental results highlight a profound engineering trade-off:

- **Supervised Learning**: Offers high local precision and noise immunity, but is highly fragile. It relies on **Shortcut Learning**—using background noise as a crutch rather than learning the physical features of the machine.
- **Unsupervised Learning**: Exhibits lower local precision under noise, but is “intellectually honest.” It models the physical structure of the sound, preventing false positive feedback loops on unseen domains.

Furthermore, we highlight a critical distinction in our evaluation compared to prior work. The original MIMII benchmark [1] evaluates anomalous sound detection by training separate, isolated models for each individual machine ID (Mono-ID) and reporting the average of these independent runs. In contrast, our study introduces a joint “Multi-ID” training paradigm, where a single model must learn the normal states of multiple physical units (00, 02, 04) simultaneously. This is a mathematically and physically different problem: a joint model is exposed to intra-class domain-shift (variances in background noise, wear, and installation biases between units), which naturally degrades performance (the drop to 0.82 AUC) but represents a far more realistic industrial fleet monitoring scenario.

Standard reconstruction-based CAEs are also known to suffer from edge-frame prediction errors, particularly on complex signals. This has driven research into alternative structures, such as the Interpolation Deep Neural Network (IDNN) [3], which bypasses edge prediction by reconstructing masked center frames, suggesting a promising path for enhancing baseline reconstruction stability.

CONCLUSIONS

This comparative study establishes a rigorous, benchmarked evaluation of unsupervised and supervised deep learning paradigms for industrial acoustic anomaly detection. While our specialized “Mono-ID” CAE successfully reproduces the 0.99 AUC benchmark reported in the original MIMII dataset paper [1] for the single-machine setup, we introduce a crucial joint “Multi-ID” paradigm. This collective model trained on multiple physical units reveals a drop to 0.82 AUC, exposing the “Cost of Scalability” caused by absorbing intra-class domain-shifts (ambient and installation variances) between different physical units—a realistic industrial constraint that separate, independent models averaged in prior work [1] fail to capture.

Furthermore, our zero-shot generalization test on the unseen Machine ID 06 exposes the extreme fragility of supervised classification under physical domain shifts. Although the CNN achieved a perfect 1.00 AUC in the source training domain, it suffered a catastrophic collapse to 0.21 AUC (total class inversion) in the target domain. This proves that the supervised network relied on **Short-cut Learning**—memorizing the stationary background noise of the training environments as a convenient statistical shortcut to minimize cross-entropy loss rather than learning the physical acoustics of mechanical failure, validating the environmental domain shift warnings formalized in the MIMII DUE framework [2].

Ultimately, in Industry 4.0 fleet expansions, collecting labeled failure data for every machine unit is logistically and economically unfeasible. Our study demonstrates that local supervision creates a fragile illusion of perfect performance that collapses during domain shifts, posing a severe operational hazard. Conversely, the unsupervised CAE, despite experiencing a performance drop under heavy noise (0.71 AUC), maintained logical consistency (0.48 AUC) on the unseen domain without experiencing class inversion. Thus, modeling the underlying physical structure of healthy machine operations via unsupervised reconstruction remains the only conceptually honest, scalable, and safe path for automatic facility inspection.

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